

CMT 442 Module 1: Foundations of Neural Networks & AI

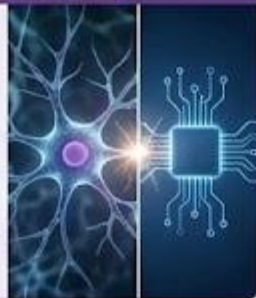
(10-Slide Summary)

Slide 1: Title & Introduction

CMT 442: Neural Networks

Module 1 - Foundations of Neural Networks and AI

Instructor: Dr. Stanley Mwangi Chege
[University Name]



Slide 2: Module Overview

From Biology to Code



History of AI



Biological Inspiration



Neural Network Basics

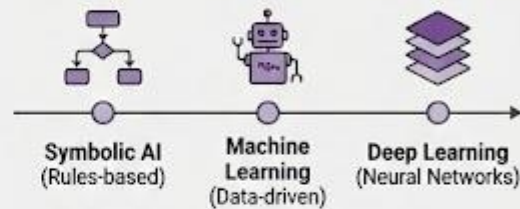
Bridging biology and computer science.

Slide 3: Learning Outcomes (CBET)

By the end of this module, you will:

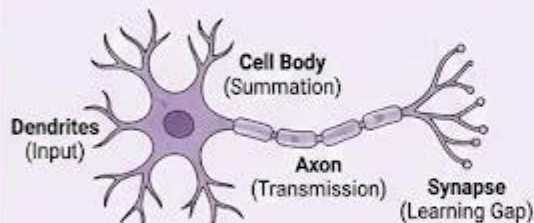
- ✓ Trace evolution of AI.
- ✓ Define basic neural network components.
- ✓ Explain biological vs. artificial similarities.
- ✓ Apply ISO/IEC 42001 governance principles.

Slide 4: The History of AI



Paradigm shift: Learning rules from data.

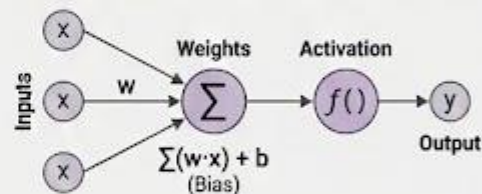
Slide 5: Biological Inspiration



Intelligence emerges from connections.

Slide 6: Neural Network Basics

Artificial Neuron (Perceptron Model)



The heart of the course.

Slide 7: Standard Mapping – ISO/IEC 42001:2023



Context & Principles



Defining AI boundaries



Context: High Risk vs. Low Risk



Relevance: Govern what you define

Slide 8: Standard Mapping – NIST AI RMF



Culture & Diversity



Policies before code

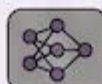


Risk Management

Slide 9: Discussion – The “Black Box” Problem



Traditional Code:
Transparent



Neural Networks:
Opaque, Millions of Weights

❓ Can you explain the decision?

Critical for Explainability & Governance.

Slide 10: Summary & Next Steps

- ✓ Mimics biological neurons
- ✓ Perceptron is basic unit
- ✓ Use ISO/IEC 42001 & NIST AI RMF

Next Module: Mathematical Models and Learning Algorithms



Assignment: Research an AI Failure due to lack of governance.

Here is a 10-slide presentation structure for **CMT 442 Module 1: Foundations of Neural Networks and AI**. This presentation establishes the theoretical and biological roots of the course while introducing the governance standards immediately.

Slide 1: Title Slide

- **Title:** CMT 442: Neural Networks
 - **Subtitle:** Module 1 – Foundations of Neural Networks and AI
 - **Instructor:** Dr. Stanley Mwangi Chege
 - **Institution:** [Your University Name]
 - **Visual:**
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Slide 2: Module Overview

- **Theme:** From Biology to Code
- **Key Topics:**
 - The History of AI: From Logic to Learning ¹
 - Biological Inspiration: How the Brain Works ²
 - Neural Network Basics: Architecture of an Artificial Neuron ³
- **Standards Focus:**
 - ISO/IEC 42001:2023 (AI Context) ⁴
 - NIST AI RMF (Govern Function) ⁵

Speaker Notes: "Welcome to CMT 442. We are about to bridge the gap between biology and computer science. In this module, we explore how scientists looked at the human brain and asked, 'Can we build this mathematically?'"

Slide 3: Learning Outcomes (CBET)

- **By the end of this module, you will be able to:**
 1. **Trace** the evolution of AI and the emergence of neural networks.
 2. **Explain** the structural similarities between biological neurons and artificial perceptrons.
 3. **Define** the basic components of a neural network (Inputs, Weights, Biases).
 4. **Apply** foundational governance principles from ISO/IEC 42001 to AI development⁶.

Speaker Notes: "Our goal isn't just to code; it's to understand the *nature* of the intelligence we are building. This aligns with our course objective to 'Apply responsible AI and governance principles'"

Slide 4: The History of AI

- **The Eras of AI:**
 - **Symbolic AI (Good Old-Fashioned AI):** Rules-based, explicit programming (If-Then-Else).
 - **Machine Learning (ML):** Systems that learn from data rather than explicit rules.
 - **Deep Learning (DL):** Neural networks with many layers (the focus of this course)⁷.
- **Why Now?** The explosion of Big Data and GPU computing power.

Speaker Notes: "For decades, AI was about writing rules. Neural networks changed the paradigm: we don't write the rules; the network *learns* the rules from the data."

Slide 5: Biological Inspiration

- **The Human Neuron:**
 - **Dendrites:** Receive signals from other neurons.
 - **Cell Body (Soma):** Sums up the incoming signals.
 - **Axon:** Transmits the signal if the threshold is reached.
 - **Synapse:** The gap where chemical signals pass (learning happens here).
- **The Key Takeaway:** Intelligence emerges from the connection of many simple units⁸.

Speaker Notes: "Nature is the ultimate engineer. Our artificial networks are simplified models of this biological miracle. The 'weights' in our code are the 'synapses' in our brain."

Slide 6: Neural Network Basics (The Artificial Neuron)

- **The Perceptron Model:**
 - **Inputs (\$x\$):** The data features.
 - **Weights (\$w\$):** The strength/importance of each input.
 - **Bias (\$b\$):** The threshold to activation.
 - **Output (\$y\$):** The decision (0 or 1, Yes or No).
- **The Formula:** $y = f(\sum(w \cdot x) + b)$

Speaker Notes: "This formula is the heart of the entire course. It looks like simple algebra, but when you stack millions of these together, you get ChatGPT."

Slide 7: Standard Mapping – ISO/IEC 42001:2023

- **Standard:** ISO/IEC 42001 (Artificial Intelligence Management Systems)
- **Focus: Context and Principles** ⁹
 - **Defining AI:** Understanding the boundaries of the system.
 - **Context:** Is this AI being used for healthcare (High Risk) or a movie recommendation (Low Risk)?
 - **Relevance:** You cannot govern what you cannot define.

Speaker Notes: "We are introducing ISO 42001 on day one because you must understand the *context* of your AI. A neural network designed for a self-driving car has different requirements than one for a spam filter."

Slide 8: Standard Mapping – NIST AI RMF

- **Framework:** NIST AI Risk Management Framework (RMF)
- **Focus Function: GOVERN** ¹⁰
 - **Culture:** Cultivating a culture of risk management.
 - **Policies:** Establishing rules for how AI is built *before* you write the code.
 - **Diversity:** ensuring diverse teams build the models to reduce bias.

Speaker Notes: "The 'Govern' function tells us that safety starts at the top. Before we train a model (Module 2 & 3), we must have governance in place to handle the risks."

Slide 9: Discussion – The "Black Box" Problem

- **The Challenge:**
 - Traditional Code: We can read lines 1-100 and know exactly why an error occurred.
 - Neural Networks: The decision is made by millions of floating-point numbers (weights).
- **Question:** If a neural network denies a loan application, can you explain *why* to the customer?
- **Link to Governance:** This is why "Explainability" (Module 5) and "Governance" (Module 6) are critical.

Speaker Notes: "This is the central tension of the course: Neural networks are powerful but opaque. As we build them, keep asking: 'Do I know why it made that decision?'"

Slide 10: Summary & Next Steps

- **Recap:**

- Neural networks mimic biological neurons¹¹.
 - The basic unit is the perceptron (Inputs + Weights + Bias)¹².
 - We use ISO/IEC 42001 and NIST AI RMF to define context and govern risk¹³.
- **Next Module:** Mathematical Models and Learning Algorithms (The Math behind the Magic)¹⁴.
- **Assignment:** Research one "AI Failure" caused by a lack of governance.

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Quiz: CMT 442 Module 1 – Foundations of Neural Networks and AI

1. What is the primary title of Module 1 in the CMT 442 course outline?
 - A. Mathematical Models and Learning Algorithms
 - B. Foundations of Neural Networks and AI
 - C. Multilayer Networks and Deep Learning
 - D. AI Ethics, Governance, and Responsible AI
2. Which of the following is listed as "Indicative Content" for Module 1?
 - A. Gradient descent
 - B. Backpropagation
 - C. Biological inspiration
 - D. CNNs and RNNs
3. Which specific ISO standard is mapped to Module 1 for "AI context and principles"?
 - A. ISO/IEC 27001
 - B. ISO/IEC 42001:2023
 - C. ISO 22301
 - D. ISO 31000
4. The NIST AI Risk Management Framework (RMF) function mapped to Module 1 is:
 - A. Measure
 - B. Manage
 - C. Govern
 - D. Map
5. Which topic is NOT covered in Module 1?
 - A. History of AI
 - B. Neural network basics
 - C. Backpropagation
 - D. Biological inspiration
6. Module 1 introduces "Neural network basics." Based on the course structure, the mathematical rigor (e.g., Activation functions, Gradient descent) is primarily reserved for:
 - A. Module 2
 - B. Module 4
 - C. Module 5

D. Module 6

7. One of the Course Objectives (CBET) introduced in the foundations is to:

- A. Reverse engineer malware
- B. Apply responsible AI and governance principles in neural network development
- C. Conduct network penetration tests
- D. Perform mobile forensics

8. In the context of "Biological inspiration" (Module 1), the course draws a parallel between artificial neural networks and:

- A. The human immune system
- B. The human nervous system (brains/neurons)
- C. Genetic evolution
- D. Social ant colonies

9. Why is the "Govern" function of the NIST AI RMF introduced in Module 1?

- A. Because governance is the last step in development.
- B. Because it deals with hardware specifications only.
- C. Because establishing a culture of risk management and policy is a foundational first step.
- D. Because it is required for mobile device management.

10. Which key reference listed in the outline is most relevant to the "Deep Learning" aspect of the course history?

- A. Stallings, W. (Cryptography)
- B. Hall, J. A. (IT Auditing)
- C. Goodfellow, I., Bengio, Y., & Courville, A. (Deep Learning)
- D. Weber, R. (IS Control)

Answer Key and Explanations

1. **B** – The title of Module 1 is "Foundations of Neural Networks and AI"¹.
2. **C** – "Biological inspiration" is explicitly listed under Indicative Content for Module 1². (Gradient descent is Module 2, Backpropagation is Module 3, CNNs are Module 4).
3. **B** – "ISO/IEC 42001:2023 - AI context and principles" is the mapped standard for Module 1³.
4. **C** – "NIST AI RMF - Govern" is the specific function mapped to Module 1⁴.
5. **C** – "Backpropagation" is covered in Module 3 (Multilayer Networks and Deep Learning)⁵.

6. **A** – "Mathematical Models and Learning Algorithms" (including activation functions and gradient descent) are the focus of Module 2 ⁶.
7. **B** – The course objective is to "Apply responsible AI and governance principles in neural network development"⁷.
8. **B** – The topic "Biological inspiration" ⁸ refers to the structural mimicry of biological neurons in the brain.
9. **C** – The "Govern" function is mapped to the "Foundations" module⁹, implying that policy and risk culture are foundational prerequisites before technical development begins.
10. **C** – Goodfellow, Bengio, & Courville (2016) *Deep Learning* is the key reference for the deep learning content¹⁰.